# Automatic multilevel threshold using open-polygon search with mean and bin distance as stop condition

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hresholding is a specific case of contrast stretching where the image intensities are clipped into binary values in terms of the image histogram [1]. By extension, the multilevel thresholding is a specific case of contrast stretching where the image is divided into several regions which are then thresholded separately, this processing helps in image analysis enhancing region segmentation and object detection.

## 1 Introduction

Thresholding is a common technique for segmentation in images, which is useful in different contexts where it is intended any object detection, contour detection, object detection, etc. Here, the main proposal is to generate a program which can automatically detect the maxima of an image histogram and then processing with a multilevel threshold. For this approach, it will be used an open-polygon search method which will be explained in the next section.

## 2 Methods

In this section, it will be explained the different processes needed to achieve the proposed method.

## 2.1 Histogram

First of all, it is good to construct the histogram of the image which is made by:

$$H = \left[ \sum_{j=0}^{n-1} \sum_{i=0}^{m-1} \phi_l(A_{ij}) \right]_{l=0}^{255} \tag{1}$$

where

$$\phi_l(A_{ij}) = \begin{cases} 1 & \text{if } l = A_{ij} \\ 0 & \text{if } l \neq A_{ij} \end{cases} \tag{2}$$

## 2.2 Maxima search

For any multilevel thresholding is needed to specify the levels of reference, so in this section it'll be explained the open-polygon search method.

**Algorithm 1** Complete Histogram Maxima Detection

**Require:** Histogram $H[0..L-1]$, Window size $s$
**Ensure:** Binary maxima array $M[0..L-1]$, Count $N_{\text{maxima}}$

Initialize $M[i] \leftarrow$ false for $i = 0$ to $L-1$
$N_{\text{maxima}} \leftarrow 0$, $i \leftarrow 0$
**while** $i < L$ **do**
  $w \leftarrow \min(s, L-i)$
  $(found, pos) \leftarrow \text{ConstrainedMaxima}(H, i, w)$
  **if** $found$ **then**
    $M[i + pos] \leftarrow$ true
    $N_{\text{maxima}} \leftarrow N_{\text{maxima}} + 1$
    $i \leftarrow i + w$ ▷ Skip processed polygon
  **else**
    $i \leftarrow i + 1$ ▷ Continue search
  **end if**
**end while**
**return** $M, N_{\text{maxima}}$

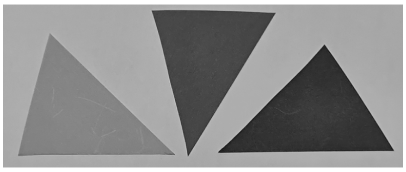
**function** CONSTRAINEDMAXIMA($H$, $start$, $width$)
  **if** $width = 0$ **then return** (false, 0)
  **end if**
  $\mu \leftarrow \frac{1}{width} \sum_{j=0}^{width-1} H[start + j]$ ▷ Polygon mean
  $\delta \leftarrow width/2$ ▷ Position constraint
  $maxid \leftarrow 0$
  **for** $j \leftarrow 1$ to $width - 1$ **do**
    **if** $H[start + j] > H[start + maxid]$ **then**
      $maxid \leftarrow j$
    **end if**

**end for**
$found \leftarrow (H[start + maxid] > \mu) \wedge (maxid < \delta)$
**return** $(found, maxid)$
**end function**

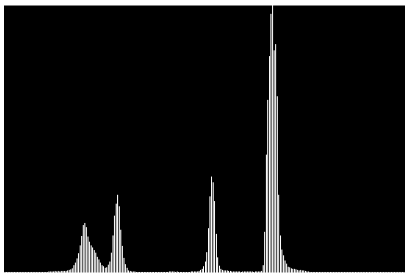
The search for the test images were made with a width of 80

## 3 Results

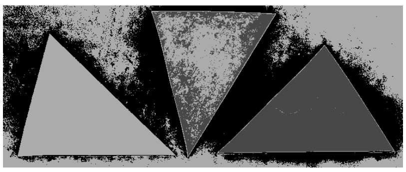
The results obtained where a full automatic multilevel thresholding.



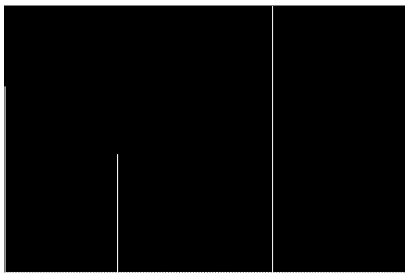
(a) Test 1: Original



(b) Test 1: Original Histogram

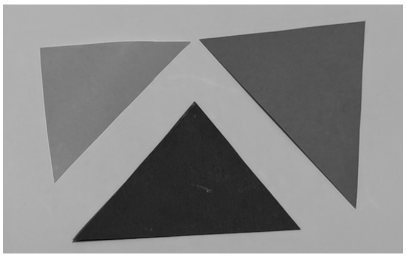


(c) Test 1: Thresholded



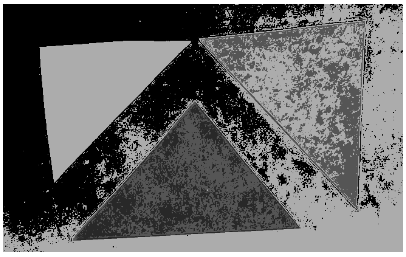
(d) Test 1: Thresholded Histogram

**Fig. 1**: Comparative analysis of thresholding results for HistoTest01.bmp



(a) Test 2: Original

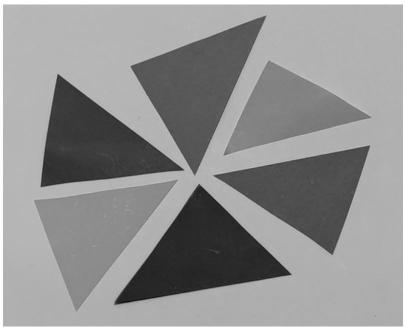
(b) Test 2: Original Histogram



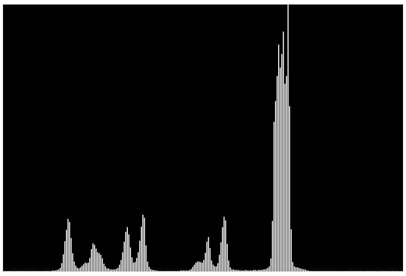
(c) Test 2: Thresholded

(d) Test 2: Thresholded Histogram

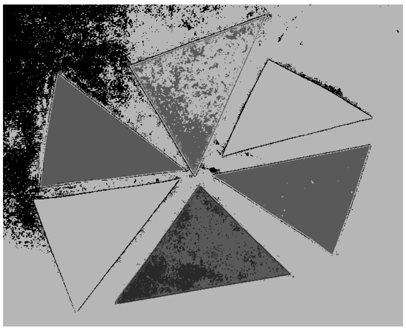
**Fig. 2**: Comparative analysis of thresholding results for HistoTest02.bmp



(a) Test 3: Original



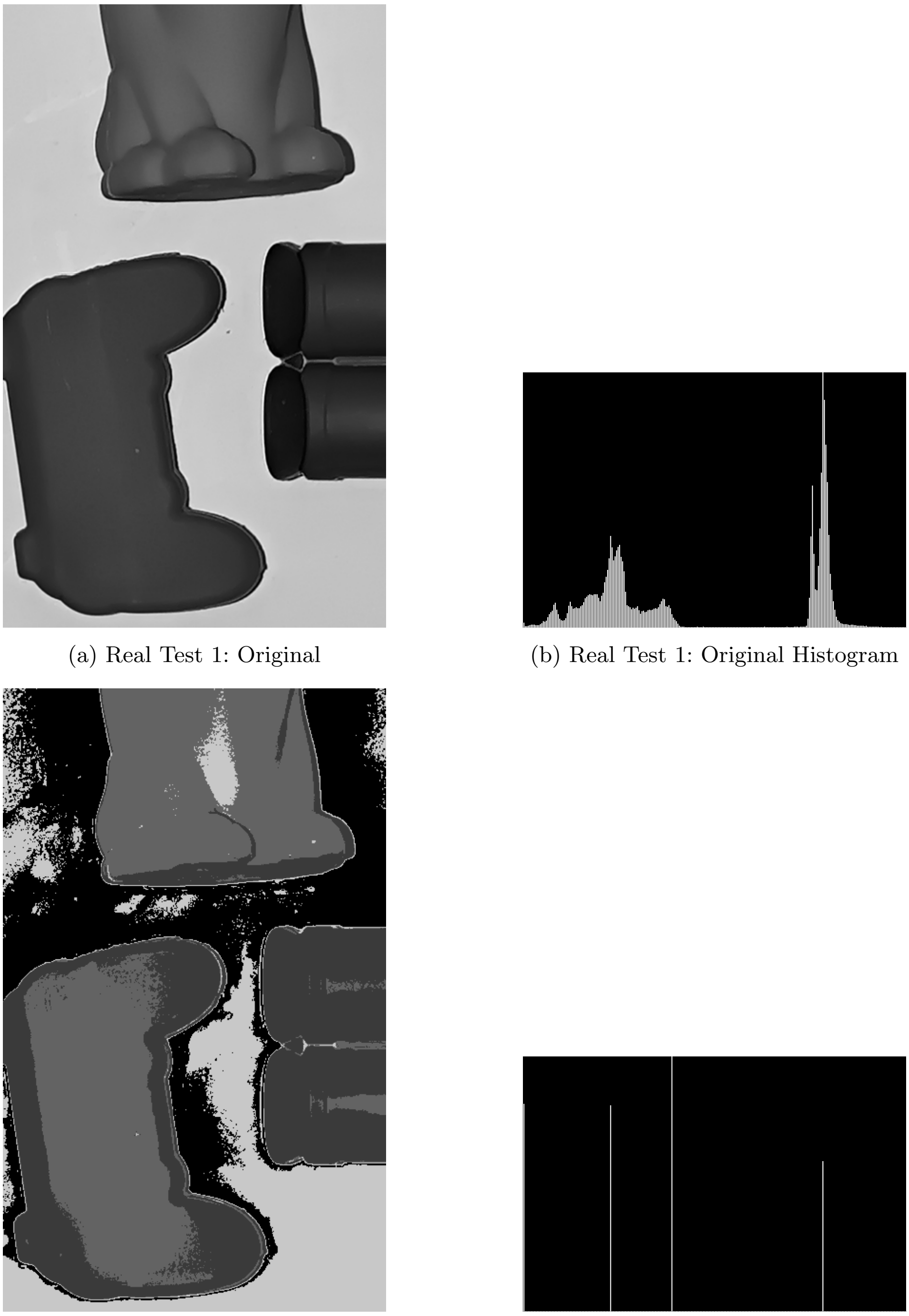
(b) Test 3: Original Histogram



(c) Test 3: Thresholded

(d) Test 3: Thresholded Histogram

**Fig. 3**: Comparative analysis of thresholding results for HistoTest03.bmp



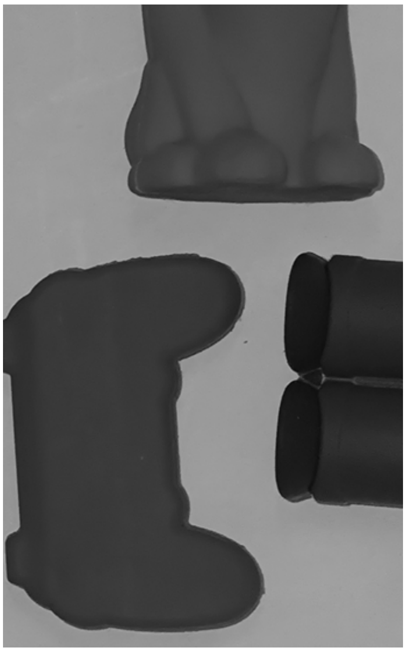
(a) Real Test 1: Original

(b) Real Test 1: Original Histogram

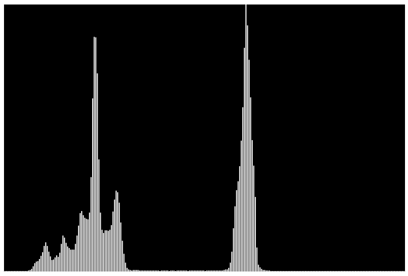
(c) Real Test 1: Thresholded

(d) Real Test 1: Thresholded Histogram

**Fig. 4**: Comparative analysis of thresholding results for HistoTestReal01.bmp

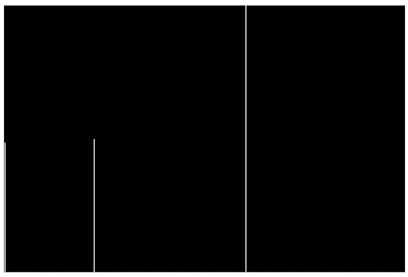


(a) Real Test 2: Original



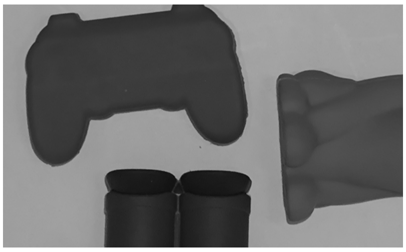
(b) Real Test 2: Original Histogram

(c) Real Test 2: Thresholded

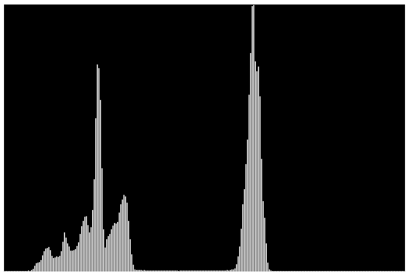


(d) Real Test 2: Thresholded Histogram

**Fig. 5**: Comparative analysis of thresholding results for HistoTestReal02.bmp

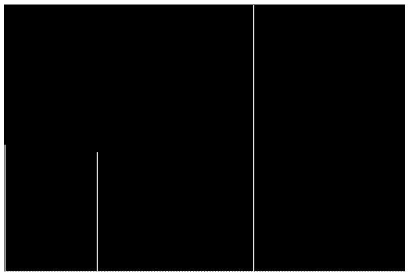


(a) Real Test 3: Original



(b) Real Test 3: Original Histogram

(c) Real Test 3: Thresholded



(d) Real Test 3: Thresholded Histogram

**Fig. 6**: Comparative analysis of thresholding results for HistoTestReal03.bmp

# 4 Discussion

The code is fully functional which can be useful in different contexts, however, there is as in every tool cases where it can perform better than others. The main point is to search a bin distance which can be optimum for each image and light conditions.

# 5 Conclusion

In this work it was proposed a method to automatically detect the maxima of an image histogram and then processing with a multilevel threshold which can be useful in different contexts where it is intended any object detection, contour detection, object detection, etc.

The main goal was achieved, but regarding that the bin distance for this approach should be tuned depending on the image needs and features.

It can be later used in HSV space for color segmentation, and it should be useful for binarizaton by putting a width of 256.

This approach is also perfectible, so this is the first step in this project.

# References

[1] Jain, A.K.: Fundamentals of Digital Image Processing. Pearson, Upper Saddle River, NJ (1988)